

Notes: Barberis, Greenwood, Jin, and Shleifer (JFE, 2018)

Extrapolation and Bubbles

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1 Big picture

- **Question.** How can extrapolative expectations generate asset-price bubbles, bubble crashes, and the high trading volume that is historically associated with speculative episodes?
- **Motivation.** Historical accounts of bubbles emphasize that investors extrapolate recent price increases. Yet many formal bubble models either rely on rational bubbles or exogenous disagreement, and often do not explain why volume becomes unusually high during bubbles.
- **Core model.** The paper builds a finite-horizon asset-pricing model with a risky asset, fundamental traders, and extrapolators. Extrapolators put weight on two signals: a growth signal based on past price changes and a value signal based on overvaluation relative to fundamentals.
- **Key new ingredient.** Extrapolators “waver” over time in the relative weight they place on growth versus value. This small random shifting of attention generates heterogeneous trading even when investors share the same public information.
- **Bubble mechanism.** A sequence of good cash-flow news raises prices. Extrapolators then infer further price increases from past returns and buy aggressively. Fundamental traders initially sell to them, but once the asset is sufficiently overvalued, fundamental traders hit short-sale constraints and exit.
- **Volume mechanism.** During bubbles, the growth signal is strongly positive and the value signal is strongly negative. Because the two signals are both large in magnitude and point in opposite directions, even small wavering generates large portfolio adjustments and high volume.
- **Distinctive empirical prediction.** Volume during a bubble should be strongly related to past returns. This is not a generic prediction of rational-bubble or disagreement models.
- **Evidence.** The paper studies four historical bubbles and the late-1990s technology bubble in more detail. It documents that volume tracks past returns, that extrapolator-like investors account for more volume as the bubble grows, and that some mutual funds and hedge funds move in and out of Internet stocks in a way consistent with wavering.
- **Economic takeaway.** Bubbles can arise from a psychologically plausible interaction between extrapolation, valuation concerns, and unstable attention. The same mechanism explains both overpricing and trading volume.

2 Incremental contribution of the paper

- **From narrative extrapolation to a formal bubble model.** The paper takes the central idea in historical accounts of bubbles - investors project past price increases into the future - and embeds it in a tractable asset-pricing model.
- **A two-signal extrapolator.** Prior extrapolation models often focus only on return extrapolation. This paper lets extrapolators also care about value, so bubble investors face an internal conflict between greed and fear.
- **Wavering as a volume engine.** The paper’s most distinctive contribution is to show how small random shifts in signal weights can produce large trading volume precisely during bubbles, without assuming that disagreement exogenously rises.

- **A theory of displacement.** The model formalizes Kindleberger’s idea that bubbles often begin with good fundamental news. The largest bubbles arise when positive shocks first build up and then taper off, because this pattern maximizes extrapolative enthusiasm.
- **A price-spiral mechanism.** When overvaluation becomes extreme, less bullish extrapolators exit and the asset is held by the most bullish extrapolators. This selection effect can amplify the bubble.
- **A bridge between behavioral expectations and volume.** The paper connects survey and experimental evidence on extrapolative expectations with the classic empirical fact that bubbles have intense trading.
- **Distinction from rational-bubble and disagreement models.** Rational bubbles have difficulty explaining volume; disagreement models often assume exogenous disagreement. Here, overpricing endogenously creates disagreement because wavering investors react differently to large conflicting signals.
- **Empirical validation of a distinctive prediction.** The paper tests the model’s prediction that bubble volume is strongly related to past returns using utility stocks in 1929, technology stocks in 1998-2000, housing in 2004-2006, and oil in 2007-2008.

3 Data and empirical setting

3.1 Model environment

The theoretical economy has dates $t = 0, 1, \dots, T$, a risk-free asset with return normalized to zero, and a risky asset with fixed supply Q . The risky asset pays a liquidating cash flow at date T :

$$D_T = D_0 + \varepsilon_1 + \dots + \varepsilon_T, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2).$$

Interpretation: The risky asset has a fundamental value based on expected final cash flow, but the market price can deviate from that value because extrapolators trade on past price changes.

3.2 Historical bubble evidence

The paper evaluates several empirical implications using four bubble episodes:

- **U.S. utility stocks in 1928-1929.** Utilities are identified using CRSP SIC codes 4900-4990 and compared with the broader market.
- **Technology stocks in 1998-2000.** The paper uses the .com stock list from Ofek and Richardson and CRSP returns and turnover.
- **U.S. housing in 2004-2006.** Prices are measured using the Case-Shiller 20-City Index, and volume is proxied by existing-home sales.
- **Oil and commodities in 2007-2008.** The paper uses United States Oil (USO), an ETF with oil exposure, with price and turnover from CRSP.

Interpretation: These episodes are not used to prove definitively that bubbles exist. They are used to test whether the model’s distinctive volume-past-return prediction is visible in well-known bubble-like episodes.

3.3 Investor-level evidence in the technology bubble

For the technology bubble, the paper also uses mutual fund holdings from Thomson Reuters and a hedge fund sample from Brunnermeier and Nagel. The goal is to study who trades during the bubble and whether investors' exposure to Internet stocks fluctuates in a way consistent with wavering.

Interpretation: This investor-level evidence is important because the model is not just a price model. It has predictions about the composition of volume and the behavior of extrapolator-like investors.

4 Models and empirical specifications

4.1 Fundamental traders

Fundamental traders demand the risky asset according to

$$N_t^F = \max \left\{ \frac{D_t - \gamma\sigma_\varepsilon^2(T-t-1)Q - P_t}{\gamma\sigma_\varepsilon^2}, 0 \right\}.$$

If all investors were fundamental traders, the equilibrium price would be

$$P_t^F = D_t - \gamma\sigma_\varepsilon^2(T-t)Q.$$

Interpretation: Fundamental traders lean against mispricing. They buy when the price is low relative to fundamental value and sell when it is high. The short-sale constraint prevents them from fully absorbing extrapolator demand in a bubble.

4.2 Extrapolative growth signal

A pure extrapolator forms a growth signal from past price changes:

$$X_t = (1 - \theta) \sum_{k=1}^{t-1} \theta^{k-1} (P_{t-k} - P_{t-k-1}) + \theta^{t-1} X_1, \quad 0 < \theta < 1.$$

A high value of X_t means recent price changes have been high, so the extrapolator expects continued appreciation.

Interpretation: The parameter θ governs the persistence of extrapolation. A lower θ puts more weight on very recent price changes, producing more intense but shorter-lived bubbles.

4.3 Value signal and two-signal demand

Extrapolators do not ignore valuation completely. Extrapolator i combines a value signal and a growth signal:

$$N_t^{E,i} = \max \left\{ w_{i,t} \left(\frac{D_t - \gamma\sigma_\varepsilon^2(T-t-1)Q - P_t}{\gamma\sigma_\varepsilon^2} \right) + (1 - w_{i,t}) \left(\frac{X_t}{\gamma\sigma_\varepsilon^2} \right), 0 \right\}.$$

The value signal is negative when the asset is overvalued. The growth signal is positive when prices have recently risen.

Interpretation: During a bubble, extrapolators face a conflict. The value signal says “sell because the asset is expensive,” while the growth signal says “buy because prices have been rising.”

4.4 Wavering

The key behavioral assumption is that the weight on value varies slightly over time:

$$w_{i,t} = w_i + u_{i,t}, \quad u_{i,t} \sim N(0, \sigma_u^2),$$

where the distribution is truncated so that the weight remains in the admissible interval.

Interpretation: Wavering captures random shifts in attention between value and growth. The degree of wavering is constant over time, but its effect on trading is much larger during bubbles because the two signals are then very large and conflicting.

4.5 Equilibrium price

When the active set of traders is I^* , the market-clearing price can be written as

$$P_t = D_t + \frac{\sum_{i \in I^*} \mu_i (1 - w_{i,t})}{\sum_{i \in I^*} \mu_i w_{i,t}} X_t - \gamma \sigma_\varepsilon^2 Q \frac{(\sum_{i \in I^*} \mu_i w_{i,t}) (T - t - 1) + 1}{\sum_{i \in I^*} \mu_i w_{i,t}}.$$

Interpretation: The price is anchored by expected fundamentals D_t , pushed upward by extrapolator demand when X_t is high, and discounted for risk bearing.

4.6 Trading volume

The model defines volume as one-half of the total absolute change in positions:

$$Volume_t = \frac{1}{2} \left(\mu_0 |N_t^F - N_{t-1}^F| + \sum_{i=1}^I \mu_i |N_t^{E,i} - N_{t-1}^{E,i}| \right).$$

Interpretation: The factor one-half prevents double counting: every purchase has a corresponding sale. High volume in the model comes from large position changes across trader types and within the group of extrapolators.

4.7 Empirical volume-past-return prediction

The central empirical prediction is:

$$Volume_{t+1} \quad \text{should be strongly increasing in} \quad P_t - P_{t-4}$$

during bubble periods.

Interpretation: Since X_t is a weighted average of past price changes, large past returns make the growth signal stronger. In a bubble, this interacts with wavering and generates high volume.

4.8 Composition of bubble volume

For technology stock i in quarter t , the paper measures whether volume is associated with extrapolator-like mutual funds:

$$Volume_Mom_{i,t} = \frac{\sum_j Buys_{i,j,t} Fundmom_{j,t-2} + \sum_j Sells_{i,j,t} Fundmom_{j,t-2}}{\sum_j Buys_{i,j,t} + \sum_j Sells_{i,j,t}}.$$

The variable $Fundmom_{j,t-2}$ measures how much fund j 's earlier portfolio was tilted toward high past-return stocks.

Interpretation: A high value of $Volume_Mom$ means trading volume is dominated by funds with extrapolator-like portfolios.

5 Identification and empirical design

5.1 What the theory identifies

The model is not a fully rational benchmark with a single departure. It identifies a specific behavioral mechanism: extrapolators combine a growth signal and a value signal, and small shifts in attention between the two signals generate large trading precisely when the signals conflict.

Interpretation: The theory's contribution is mechanism design rather than formal causal identification. It explains why bubbles and high volume should arise together.

5.2 Why good news matters

The model produces bubbles after a sequence of positive cash-flow shocks. Good fundamental news raises prices first; then extrapolative demand amplifies the price increase beyond fundamentals.

Interpretation: This formalizes the historical idea of “displacement.” A bubble does not begin from nothing; it begins from a plausible positive story that investors then extrapolate.

5.3 Three stages of a bubble

The bubble has three stages:

- **Stage 1: Fundamental traders are still in the market.** Positive news raises prices; extrapolators buy and fundamental traders sell.
- **Stage 2: Fundamental traders exit.** The asset becomes sufficiently overvalued that fundamental traders hit the short-sale constraint. Extrapolators trade among themselves because of wavering.
- **Stage 3: Fundamental traders re-enter.** As extrapolator enthusiasm fades and prices fall, fundamental traders buy from extrapolators.

Interpretation: The second stage is the key novelty for volume. Once fundamental traders are out, high volume is not simply fundamentalists versus optimists; it is trading among extrapolators who waver between greed and fear.

5.4 Comparison with rational bubbles

Rational-bubble models typically write price as

$$P_t = P_{D,t} + B_t, \quad B_t = \frac{E_t[B_{t+1}]}{1+r}.$$

Interpretation: These models do not naturally explain how bubbles begin, why volume is high, or why investor expectations are extrapolative. The paper’s model addresses these features directly.

5.5 Comparison with disagreement models

Disagreement models can explain high volume when investors disagree. The difference is that the disagreement is often exogenous. In this paper, disagreement is endogenous: the bubble makes growth and value signals large, and wavering then converts small differences in attention into large demand differences.

Interpretation: The paper does not deny the importance of disagreement. It explains why disagreement should grow during a bubble.

5.6 Empirical design and limitations

The empirical tests are suggestive rather than definitive. They ask whether distinctive patterns predicted by the model appear in historical data:

- Does volume track past returns during bubble episodes?
- Does the composition of volume shift toward extrapolator-like investors?
- Do funds move in and out of bubble assets in a way consistent with wavering?

Interpretation: The evidence supports the model’s mechanism, but the paper does not claim to settle whether every studied episode is unquestionably a bubble.

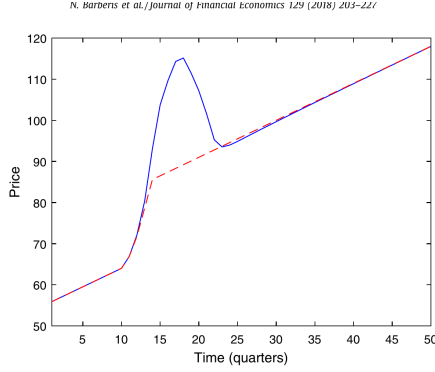
6 Figures

6.1 Figures 1 and 3: Bubble prices and the price spiral

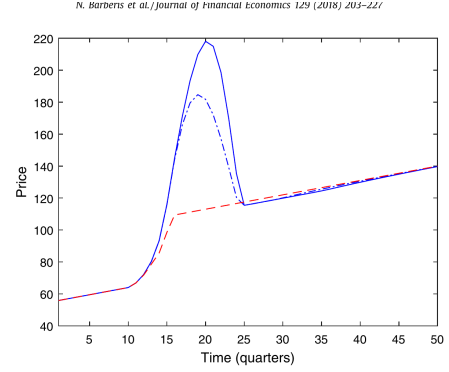
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- Figure 1 plots the price and fundamental value after a sequence of positive cash-flow shocks: 10 zero shocks, then shocks of 2, 4, 6, and 6, followed by zero shocks.
- The price rises far above fundamental value because positive news raises prices and extrapolators then increase demand based on recent price changes.
- Figure 3 uses a larger positive-shock sequence and compares a wavering economy to an economy with homogeneous extrapolators.
- The solid price line in Figure 3 rises above the homogeneous-extrapolator line, illustrating a price spiral.

Economic message: The model can generate a bubble from good news alone, but wavering can further amplify overpricing by selecting the most bullish extrapolators as asset holders.



(a) Figure 1: prices in a bubble



(b) Figure 3: price spiral

6.2 Figure 2: Comparative statics

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- Figure 2 varies four key parameters: the fraction of fundamental traders μ_0 , the average value-signal weight w , the extrapolation parameter θ , and the wavering parameter σ_u .
- Lower μ_0 produces larger bubbles because there are fewer fundamental traders to absorb extrapolator demand.
- Lower w produces larger bubbles because extrapolators care less about valuation.
- Lower θ produces more intense bubbles because recent price changes receive more weight.
- Higher σ_u generates more volume and, at higher levels, can also increase overvaluation through price spirals.

Economic message: The comparative statics show that the model's bubble and volume predictions are strongest when fundamental arbitrage is limited, extrapolation is strong, and wavering is meaningful.

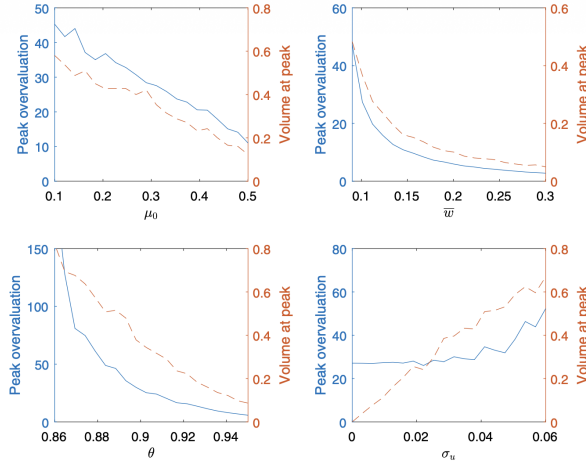


Fig. 2. Comparative statics. The four graphs correspond to four model parameters: μ_0 , $\bar{w}$, θ , and σ_u . In each graph, the solid line plots the maximum overvaluation of the risky asset across the $T = 50$ dates for the same sequence of cash-flow shocks used in Fig. 1. The dashed line plots trading volume on the date of peak overvaluation. The solid and dashed lines are computed by varying the value of the parameter on the horizontal axis while keeping the values of the other parameters at the benchmark levels specified in Section 2.

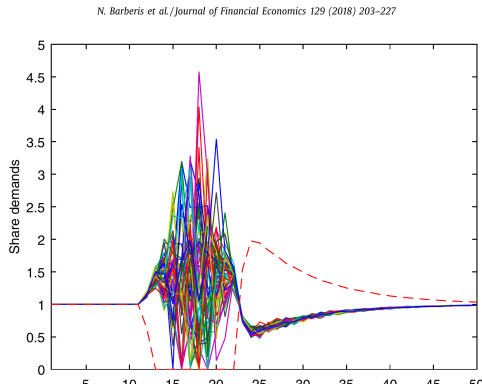
Figure 2: Comparative statics

6.3 Figures 4 and 5: Share demands and volume in a bubble

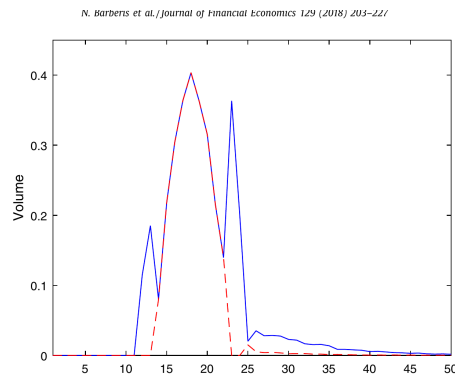
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- Figure 4 plots the demands of fundamental traders and extrapolators during the same bubble path as Figure 1.
- Extrapolator demands become highly volatile near the height of the bubble because investors waver between growth and value signals.
- Figure 5 decomposes trading volume. The model predicts three volume waves: fundamental traders selling to extrapolators, extrapolators trading among themselves, and fundamental traders re-entering as the bubble collapses.
- The middle volume wave is the distinctive one: it is generated by trading among wavering extrapolators.

Economic message: High volume is not an add-on assumption. It emerges because the bubble makes both greed and fear strong, so small shifts in attention produce large position changes.



(a) Figure 4: share demands in a bubble



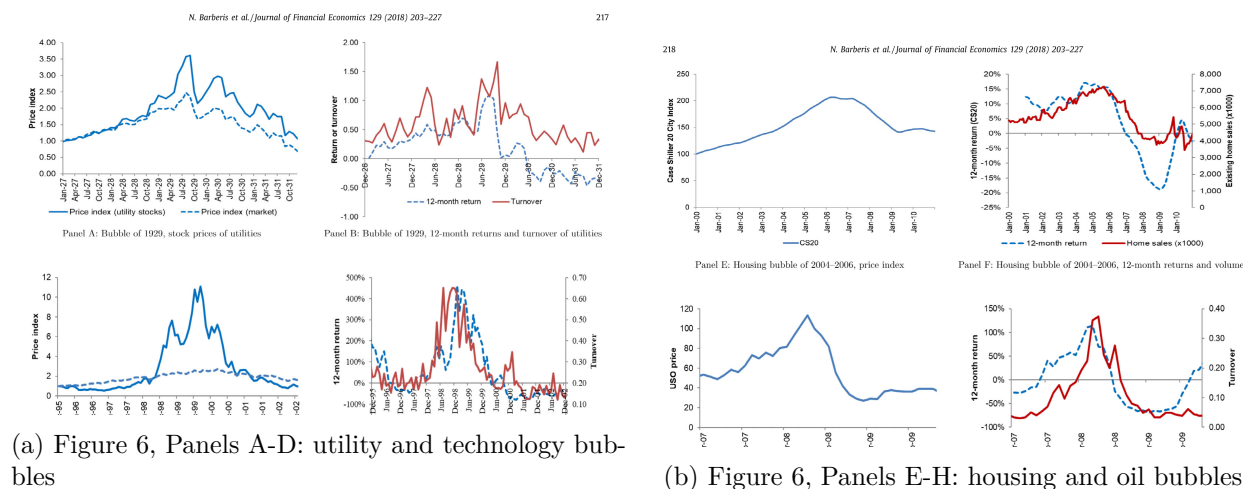
(b) Figure 5: volume in a bubble

6.4 Figure 6: Volume and past returns across historical bubbles

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- Figure 6 studies four historical bubble-like episodes: 1929 utility stocks, 1998-2000 technology stocks, 2004-2006 housing, and 2007-2008 oil.
- For each episode, the paper compares price dynamics with the relationship between past 12-month returns and turnover or sales volume.
- In all four cases, trading volume or turnover closely tracks past returns during the bubble episode.
- The correlations reported in the text are high during the bubble episodes and much lower in the post-bubble periods.

Economic message: This figure provides the paper's central empirical support: bubbles are not only high-return episodes; they are episodes where volume rises with lagged returns, as predicted by extrapolative wavering.

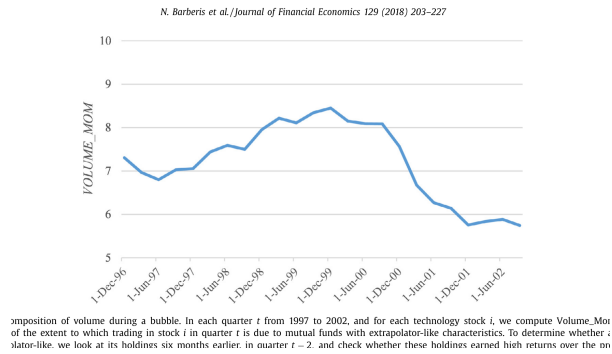


6.5 Figures 7 and 8: Composition of volume and wavering by investors

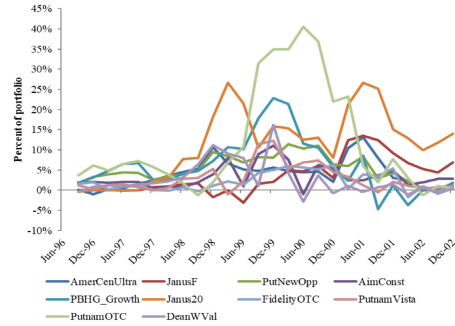
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- Figure 7 plots *Volume_Mom*, a measure of how much technology-stock trading volume is associated with mutual funds whose earlier portfolios were tilted toward high past-return stocks.
- *Volume_Mom* rises during the Internet bubble and falls after the bubble collapses.
- Figure 8 plots active changes in Internet-stock exposure for large mutual fund and hedge fund holders.
- Several investors move into, out of, and sometimes back into Internet stocks, consistent with wavering rather than a single monotone buy-and-hold decision.

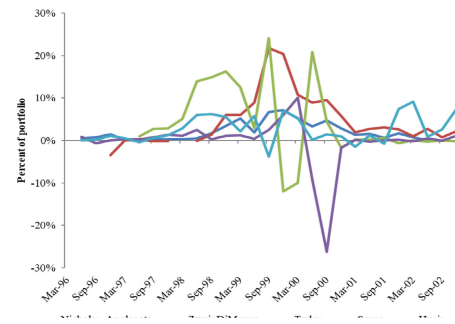
Economic message: The investor-level evidence supports the model's micro-foundation. Extrapolator-like investors become more important during the bubble, and their trading is not perfectly synchronized.



(a) Figure 7: composition of volume



Panel A: Mutual fund net purchases



(b) Figure 8: wavering by funds

7 Short summary

- This note summarizes the paper, which develops an extrapolative model of bubbles in which investors combine a past-return growth signal with a valuation signal and waver over time in the weight placed on each signal.
- Good cash-flow news can trigger a bubble because it raises prices and strengthens extrapolator demand.
- Bubbles have three stages: fundamental traders sell to extrapolators, extrapolators trade with each other, and fundamental traders re-enter as prices fall.
- Wavering is the key mechanism behind high volume at the height of the bubble.
- During normal times, wavering produces little trading because growth and value signals are small.
- During bubbles, growth and value signals are large and conflicting, so the same small wavering produces large portfolio changes.
- The model predicts that bubble volume should be strongly related to past returns.
- Evidence from utility stocks, technology stocks, housing, and oil supports this volume-past-return pattern.
- Technology-bubble evidence shows that extrapolator-like mutual funds account for more volume as the bubble develops.

- Fund-level and hedge-fund-level evidence suggests that some investors waver by moving in and out of Internet stocks.

8 Conclusion

8.1 Core conclusion

Barberis, Greenwood, Jin, and Shleifer develop a behavioral model in which extrapolation generates bubbles and wavering generates high volume. The model shows how good fundamental news can become the seed of a large overvaluation once investors use past returns as a growth signal. It also explains why bubbles are accompanied by intense trading: investors face powerful but conflicting signals and shift attention between them.

8.2 Economic intuition

The central intuition is the interaction between two forces. First, extrapolation converts past price increases into optimistic expected returns. Second, valuation concerns do not disappear; they create fear that the asset may be overpriced. During ordinary periods, the growth and value signals are small, so wavering has little effect. During bubbles, both signals become large and point in opposite directions. Small changes in how investors weight the two signals then create large trades.

8.3 Incremental contribution revisited

The paper advances the bubble literature in three ways. First, it restores extrapolative expectations to the center of bubble theory. Second, it explains high trading volume endogenously through wavering, rather than assuming exogenous disagreement. Third, it produces distinctive empirical predictions and supports them using historical bubble episodes and investor-level evidence from the technology bubble.

8.4 Implications for behavioral finance

The model suggests that bubble investors should not be described simply as irrational optimists. They can be internally conflicted: attracted by recent price growth and worried by overvaluation. This conflict makes their portfolios unstable. The key behavioral friction is not just excessive optimism; it is unstable attention between incompatible signals.

8.5 Implications for empirical work

The paper's volume-past-return prediction provides a useful diagnostic for bubble-like episodes. If volume rises with past returns, this pattern is consistent with extrapolative demand interacting with wavering. The model also suggests that investor composition matters: as bubbles develop, extrapolator-like investors should account for a larger share of trading.

8.6 Limitations

The model uses a simplified finite-horizon risky asset and stylized investor types. The empirical evidence is suggestive rather than a definitive test of whether each episode is truly a bubble. The model also treats extrapolative beliefs and wavering as behavioral primitives rather than deriving them from deeper psychology. Finally, institutional stabilizers such as issuance, shorting technology, leverage constraints, and policy intervention are simplified or absent.

8.7 Takeaway

The enduring takeaway is that bubbles can be understood as episodes in which good news, extrapolation, and unstable attention reinforce one another. Prices rise because investors extrapolate past returns, and volume explodes because investors waver between greed and fear.